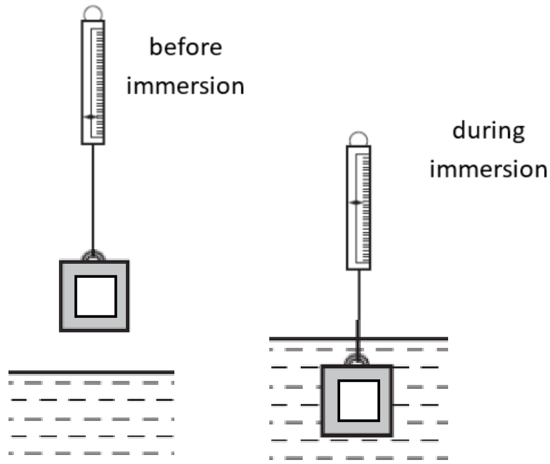
4. The diagrams show a hollow metal cube, with a sealed opening, is suspended from a spring balance before and during immersion in water. A reduction in the balance reading occurs because of the immersion.



Water is then allowed to enter the metal cube, through the opening, to fill up the empty space in the cube.

Which of the following statements is correct?

A Upthrust of the water on the cube increases, hence the balance reading will be further reduced.

B The balance reading during immersion corresponds to the upthrust of the water on the cube.

C The balance reading stays the same as the increase in weight of water in the cube is offset by the increase in upthrust of the water on the cube.

D The balance reading increases due to the increase of gravitational pull on the cube.

Ans: D

Option A: Upthrust is equal to the weight of water displaced by the cube. There is no change to the volume of the cube, hence there is no change in the upthrust.

Option B: Tension + Upthrust = Weight

Tension = Weight – Upthrust

The balance reading due to tension does not correspond to the upthrust of water on the cube.

Option C: Upthrust does not change. As water fills up the empty space in the cube, the weight of the cube will increase. Hence, balance reading will not stay the same.

Option D: Same reasoning as option C, upthrust does not change but weight increases, hence balance reading increases due to the gravitational pull (weight) on the cube.

5 Which row in the table gives the gravitational potential energy, the elastic potential